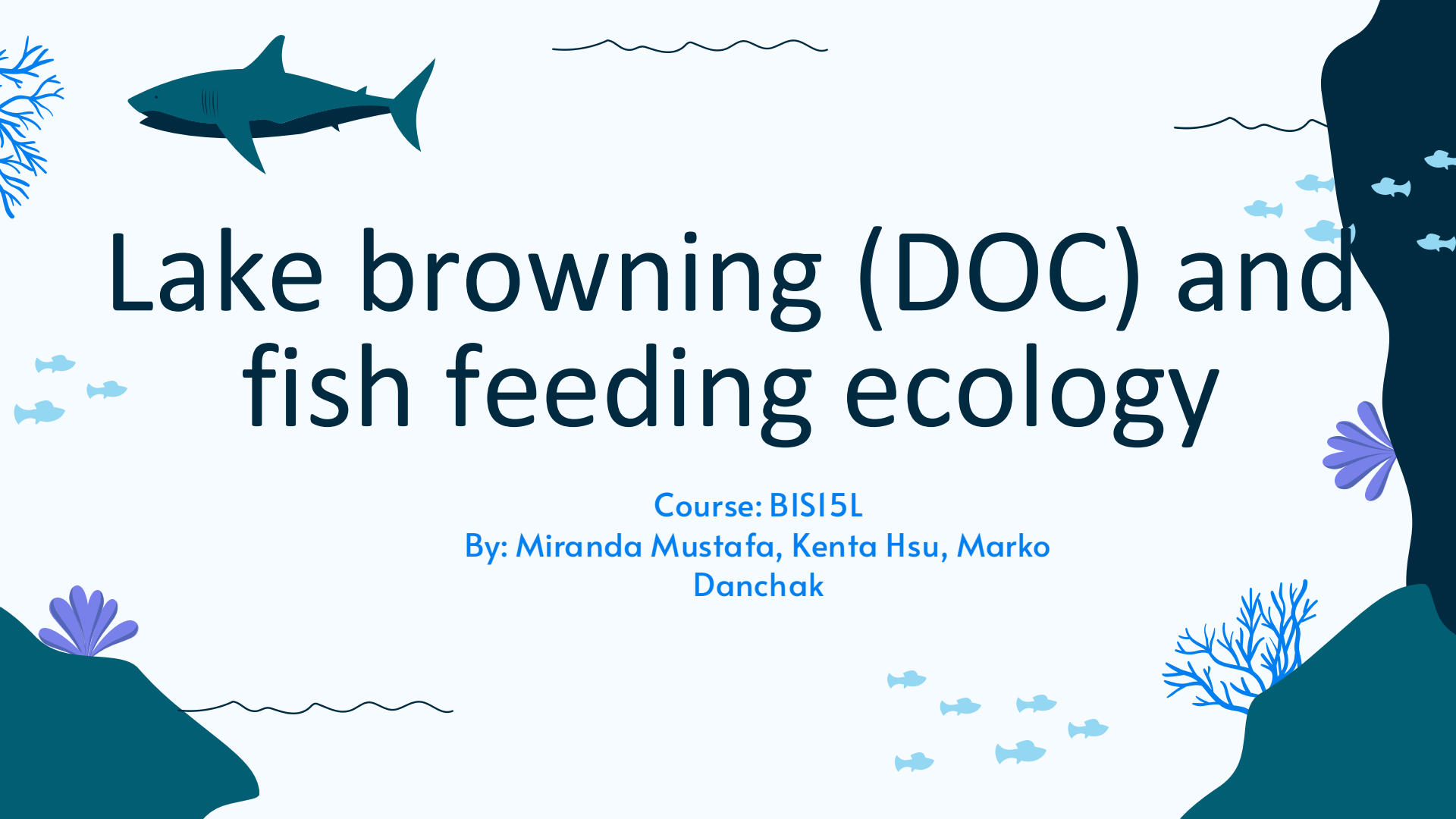


The background is a light blue gradient. In the top left, there is a large, dark teal shark swimming towards the left. To its right, there are two horizontal wavy lines representing the water surface. On the far left, there is a small, stylized blue coral branch. On the right side, there is a large, dark teal rock formation. To its left, there are several small, light blue fish swimming. At the bottom left, there is a dark teal rock formation with a small, stylized blue coral branch. At the bottom right, there is a dark teal rock formation with a small, stylized blue coral branch and several small, light blue fish swimming. The title "Lake browning (DOC) and fish feeding ecology" is written in a large, dark blue, sans-serif font in the center of the image.

# Lake browning (DOC) and fish feeding ecology

Course: BISI5L

By: Miranda Mustafa, Kenta Hsu, Marko  
Danchak



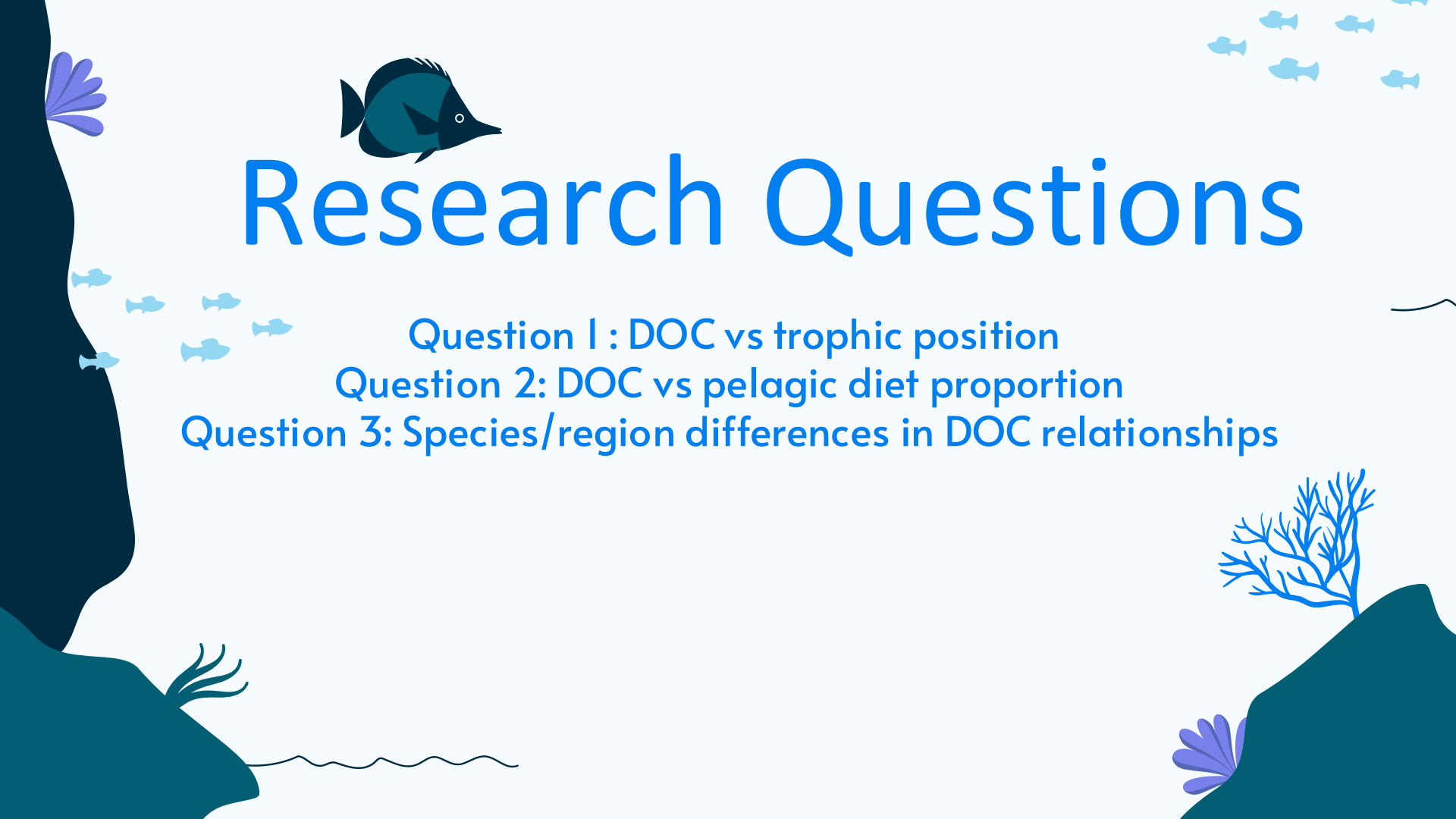
# Background

DOC or Dissolved Organic Carbon is when decaying plants or organisms causes the water to brown overtime. This makes it so the sunlight cannot penetrate through the water as easily and thus controls nutrition availability and water temperature. DOC reducing light penetration causes primary production is suppressed and reliance on algae-based food sources shifts to bacteria-based food sources



# Data Source

- Charette et al. 2024 freshwater fish dataset - -  
3,430 fish records, 78 lakes, 6 species, 6 regions
- Variables: DOC, isotopes, trophic position, pelagic  
diet proportion, fish length



# Research Questions

Question 1 : DOC vs trophic position

Question 2: DOC vs pelagic diet proportion

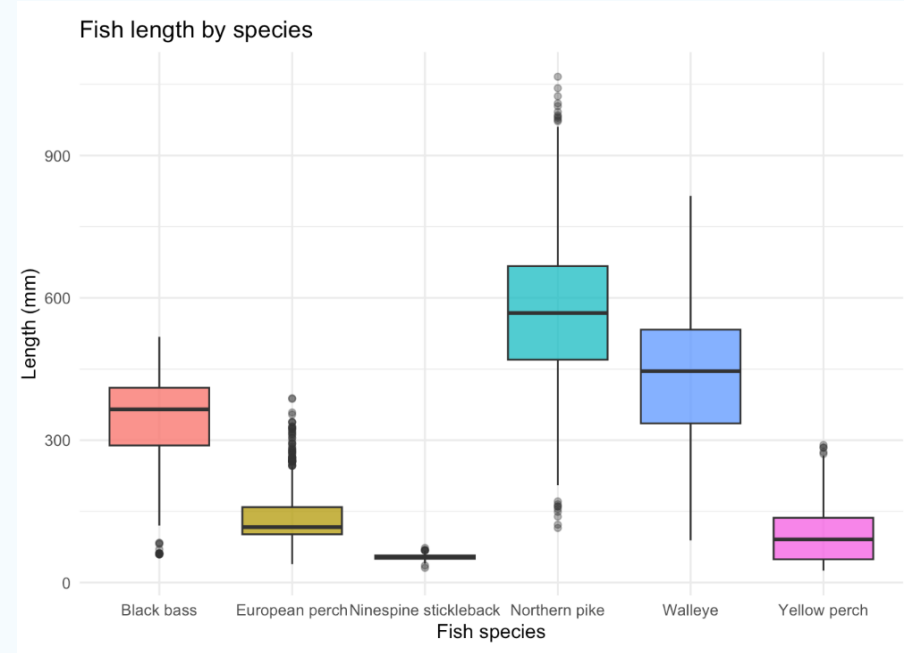
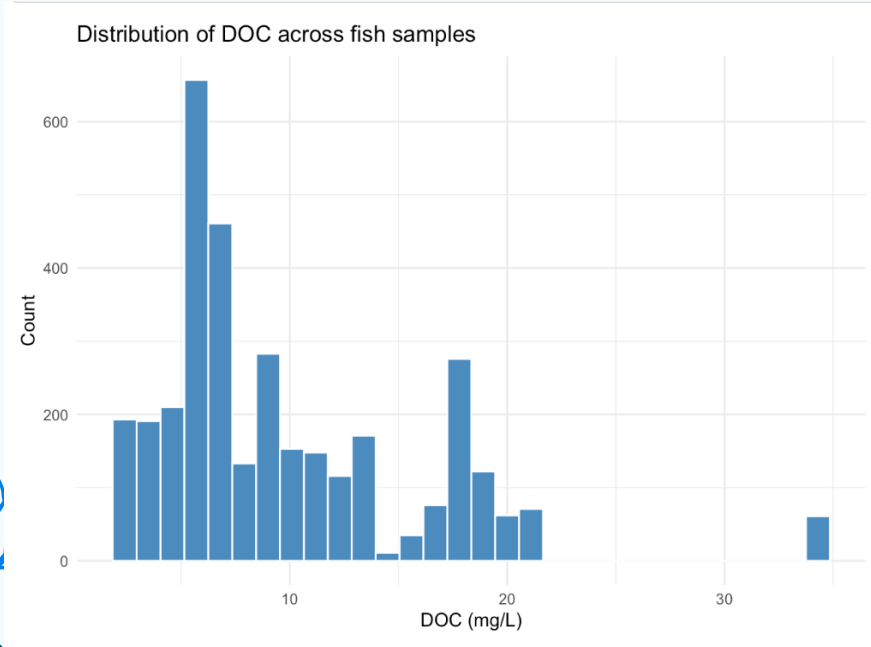
Question 3: Species/region differences in DOC relationships

# Workflow

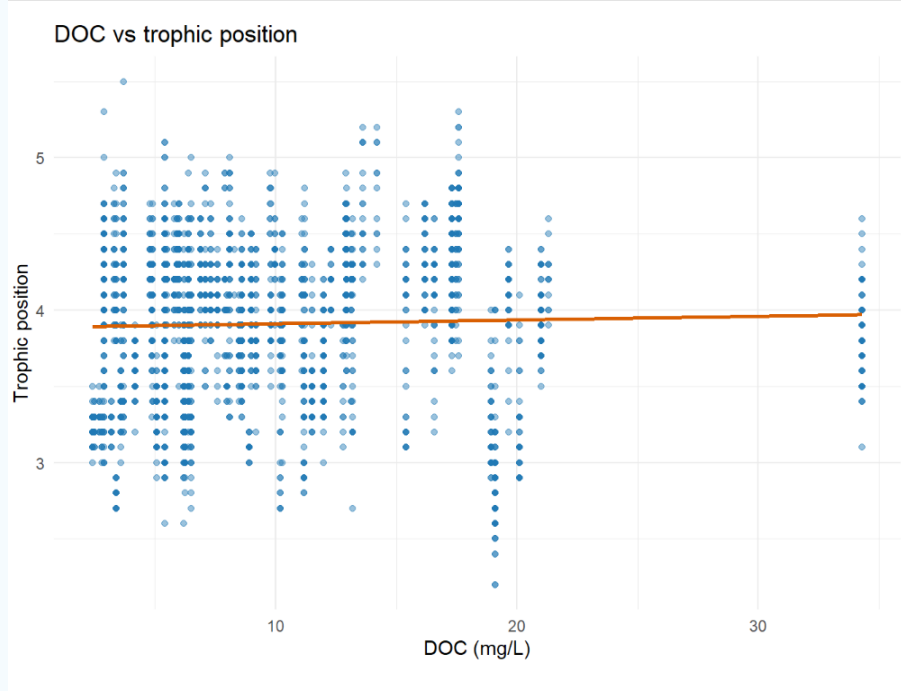
- Import and clean two CSV files
- Join fish and baseline data by lake
- Create analysis dataset and DOC categories
- Check missingness and build summaries



# Results



# Does DOC relate to trophic position?



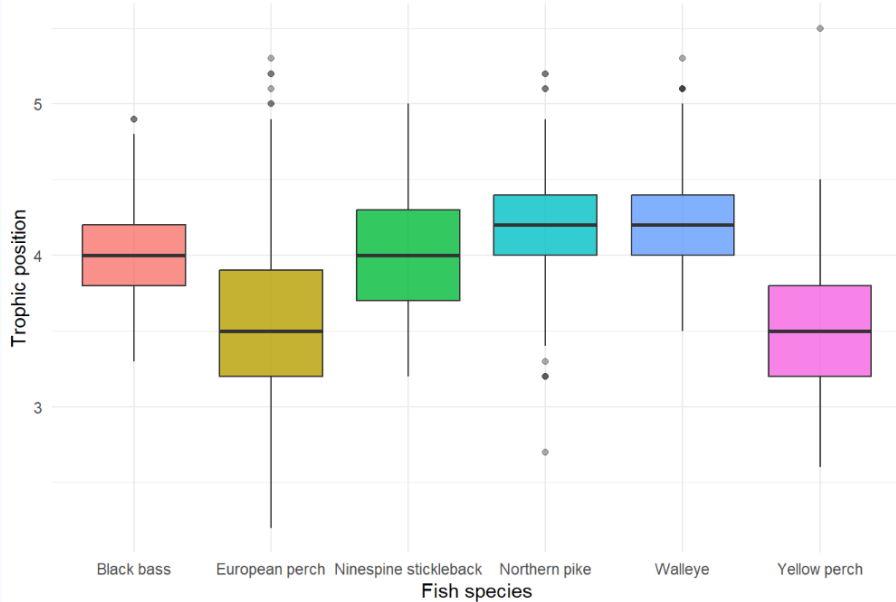
# Does DOC relate to pelagic diet



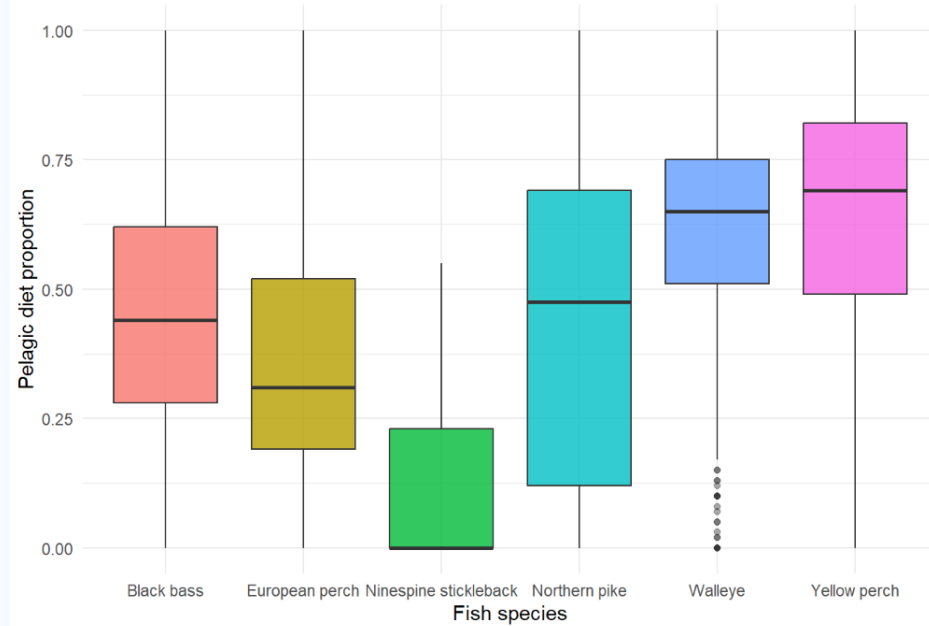


# Do DOC patterns vary by species?

Trophic position by fish species

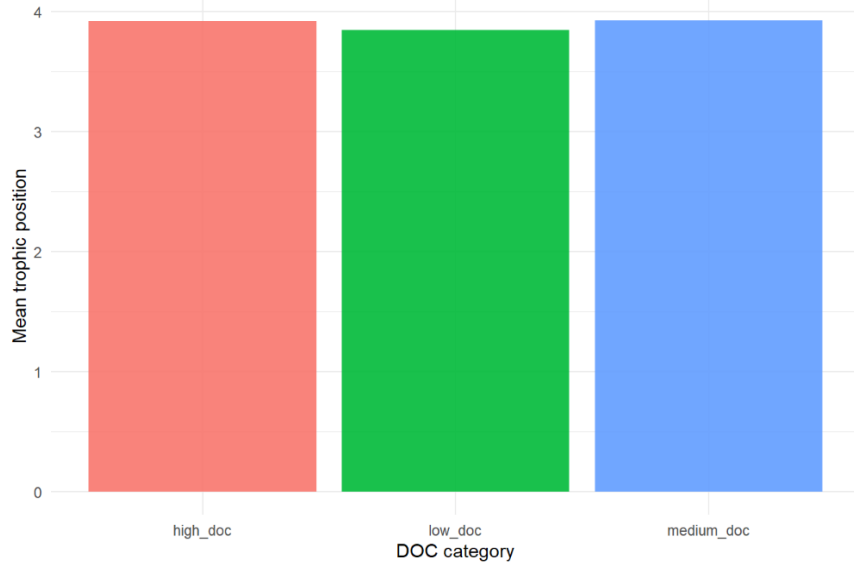


Pelagic diet proportion by fish species

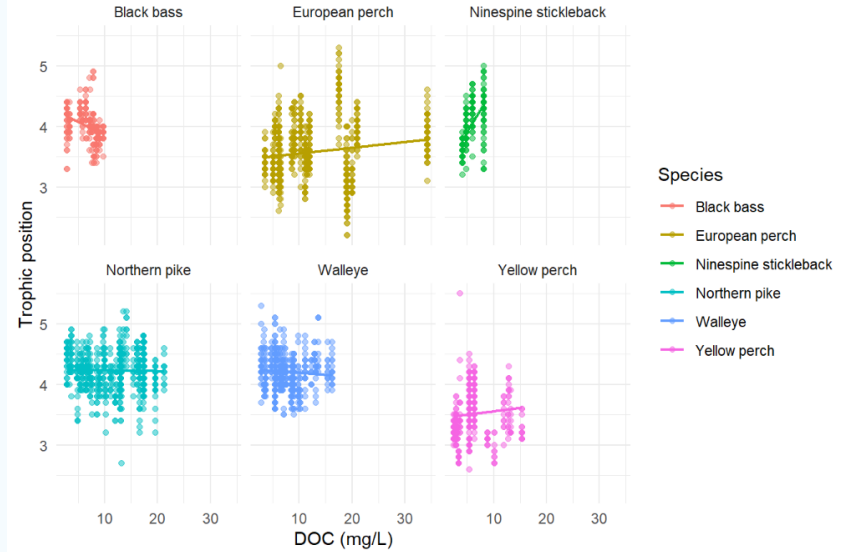


# Do DOC patterns vary by species?

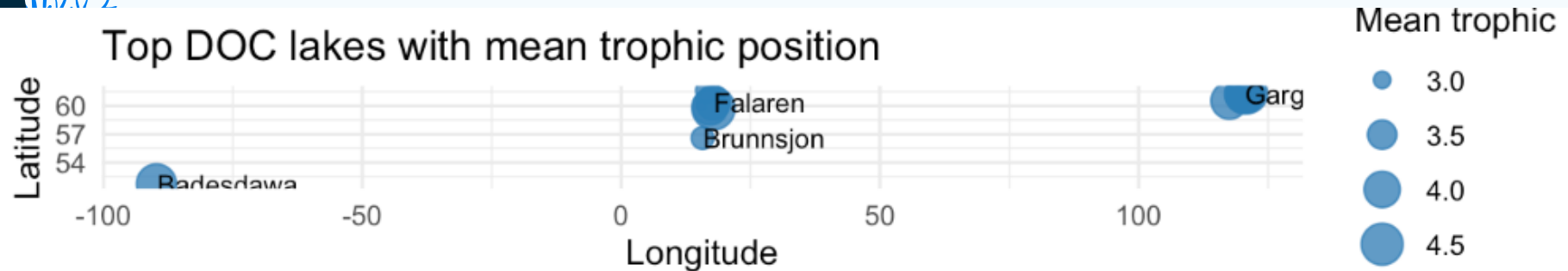
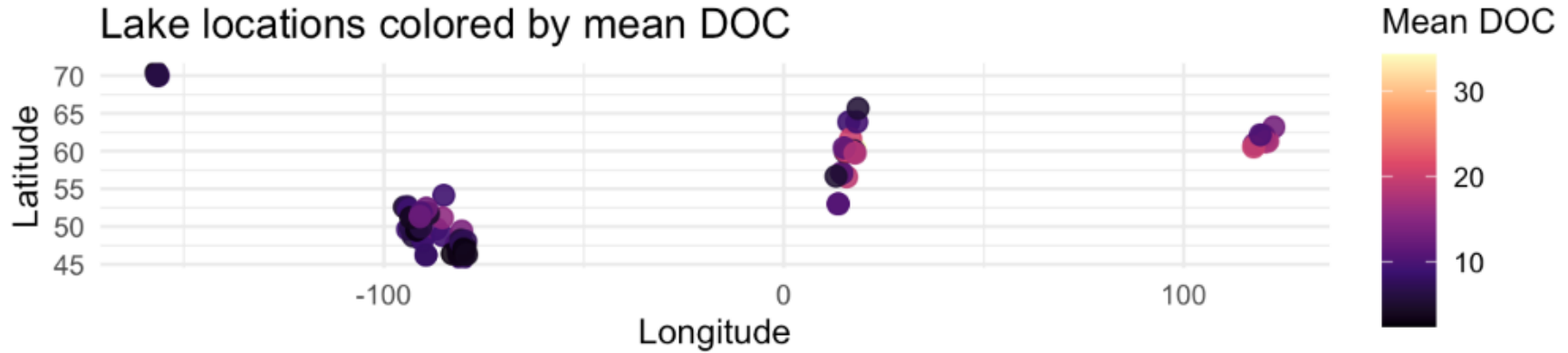
Mean trophic position by DOC category



DOC vs trophic position by species

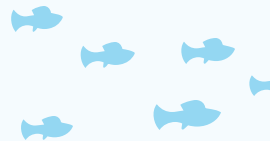
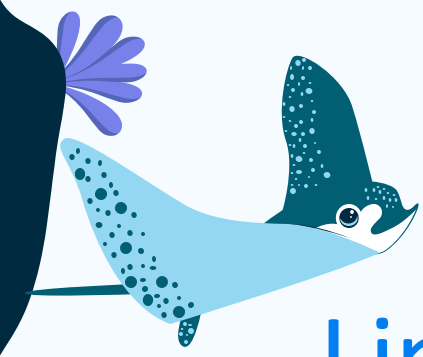


# Mapping

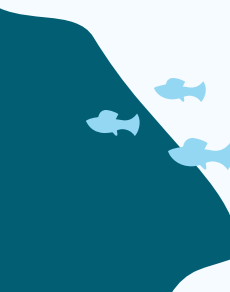


The background is a light blue gradient. It features decorative elements: a wavy line at the top left, three small blue fish in the top left, a wavy line at the top right, a blue coral-like shape on the right side, a dark blue wavy shape at the bottom left, a purple fan-like shape on the bottom right, and a group of five small blue fish at the bottom center.

# Shiny App Demo



# Limitations and Conclusion

- Overall, trophic position showed only a weak relationship with DOC.
  - But pelagic diet proportion decreased as DOC increased.
  - High DOC seem to shift **where** fish get their food from more than they shift **overall food-chain level**.
  - Species differences are also a big reason the full-dataset patterns are not simple.
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